CHAPTER 3

K-Means Clustering

Please download the sample Excel files from https://github.com/hhohho/Learn-Data-Mining-through-Excel-2. Double-click Chapter3-1a.xlsx to open it.

General Understanding

Unlike linear regression which is a predictive method, clustering is a classification method that categorizes subjects (data points) into different groups (clusters), each with a number of characteristic measurements. For example, a bank may want to categorize customers into different risk groups when considering lending loans to its customers. Such a categorizing process relies on the selected characteristics and the algorithm adopted. Pay attention to the terminologies: subject, data point, group, cluster, class, etc. In a table, a row is called a record. Since a record can be composed of multiple numerical values, it can also be called a data point. In addition, a record can contain characteristics that are used to describe somebody, so in some cases a record is also called a subject. In this book, data point, record, and subject all have the same meaning. Similarly, the three terms group, cluster, and class have the same meaning, too.

Clustering is an unsupervised data mining method. It does not require a training dataset. The two most popular types of clustering methods are partitioning clustering and hierarchical clustering. K-means clustering, where k represents the desired number of clusters, is a type of partitioning clustering. In k-means clustering, each cluster is defined by the centroid (or mean) of the data points in the cluster. For example, suppose a cluster has three data points expressed as three vectors: (1, 2, 3, 4, 5), (2, 3, 4, 5, 6), and (5, 4, 3, 2, 1). The centroid of this cluster is then ((1+2+5)/3, (2+3+4)/3, (3+4+3)/3, (4+5+2)/3, (5+6+1)/3), that is, (2.7, 3, 3.3, 3.7, 4). Note that the k-means clustering method requires all data to be numerical.

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H. Zhou, *Learn Data Mining Through Excel*, https://doi.org/10.1007/978-1-4842-9771-1_3

To start the k-means clustering process, the first task is to decide how many clusters are needed, that is, the value of k. We may need to try multiple k values to examine which k value serves our purpose the best. The second task is to select k data points as the initial centroids. We can randomly select the k centroids, or we can pick them based on the data distribution. The distances from each data point to every centroid are computed, and a data point is pushed into a cluster that is the closest to it than all other clusters are. Once every data point has been pushed into a cluster, the centroid of the points in a cluster is recomputed. The distances of each data point to every new centroid are recomputed, too, followed by categorizing points into clusters based on the shortest distance rule again. Such a process is repeated until all data points are assigned into the same clusters in two consecutive iterations, at which point the cluster centroids have stabilized and will remain the same thereafter. The process can also be stopped when the predefined maximum number of iterations has been reached.

There are different methods to compute the distances between any two data points, though Euclidean distance is commonly used. Given two data points (x1, x2, x3, x4) and (y1, y2, y3, y4), their Euclidean distance is computed as

$$\sqrt{(x1-y1)^2+(x2-y2)^2+(x3-y3)^2+(x4-y4)^2} \tag{3-1}$$

Learn K-Means Clustering Through Excel

Suppose there are 50 countries that we want to categorize into four clusters based on four attributes: murder rate, assault rate, urban population, and rape rate. All the data are stored in the Excel file Chapter3-1a.xlsx. Let's open this Excel file which has only one worksheet named 1. Yes, this worksheet is named "1". Column A stores the unique codes for each country. The data look like Figure 3-1.

	A	B	C	D	E	F	G	H	I	J	K
1											
2	Country	Murder	Assault	UrbanPop	Rape						
3	L1	12	193	51	21						
4	L2	9	222	39	42						
5	L3	6	226	72	30						
6	L4	8	177	50	17						
7	L5	7	235	86	39						
8	L6	5	170	78	37						
9	L7	0	35	75	7						
10	L8	4	198	70	15						
11	L9	13	259	74	30						
12	L10	15	136	52	24						
13	L11	5	32	82	17						
14	L12	0	54	47	12						
15	L13	10	218	79	20						
16	L14	7	26	65	19						

Figure 3-1. *Data about 50 countries*

Let's select cells G1:K1 and then click "Merge & Center". Cells G1 to K1 are merged into one cell. Enter "Mean" in this merged cell (its reference is G1). Enter other data inside cells G2:K6 as shown in Figure 3-2. Note that C1, C2, C3, and C4 are all uppercased. Be advised that there is another file named Chapter3-1a-preEntered.xlsx in which data in G1:K6 are pre-entered. You are welcome to start with Chapter3-1a-preEntered.xlsx.

	B	C	D	E	F	G	H	I	J	K
1								Mean		
2	Murder	Assault	UrbanPop	Rape		Cluster	C1	C2	C3	C4
3	12	193	51	21		Murder	2.3	5.6	11	15
4	9	222	39	42		Assault	15	53	100	200
5	6	226	72	30		UrbanPop	9.9	20	50	80
6	8	177	50	17		Rape	10	10	25	40
7	7	235	86	39						
8	5	170	78	37						
9	0	35	75	7						
10	4	198	70	15						
11	13	259	74	30						
12	15	136	52	24						
13	5	32	82	17						
14	0	54	47	12						
15	10	218	79	20						

Figure 3-2. *Set up data in G1:K6*

C1, C2, C3, and C4 represent cluster 1, cluster 2, cluster 3, and cluster 4, respectively. The numbers in cells H3:K6 are selected after a quick peek at the data ranges of each attribute. We can randomly pick some numbers for them, too. For this demonstration, let's keep these numbers exactly. Numbers in cells H3:H6, I3:I6, J3:J6, and K3:K6 stand for the centroids of C1, C2, C3, and C4, respectively.

Select cells M1:P1 and click "Merge & Center". Cells M1 to P1 are merged into one cell in which we enter the text "Distances". We are going to store the Euclidean distances from each data point to the four centroids in columns M, N, O, and P. The setup of cells M1:Q2 is illustrated in Figure 3-3.

G	H	I	J	K	L	M	N	O	P	Q
		Mean					Distances			
Cluster	C1	C2	C3	C4		C1	C2	C3	C4	Cluster
Murder	2.3	5.6	11	15						
Assault	15	53	100	200						
UrbanPop	9.9	20	50	80						
Rape	10	10	25	40						

Figure 3-3. *Setup of the cells M1:Q2*

Cell M3 stores the distance from country L1 (12, 193, 51, 21) to centroid C1 (2.3, 15, 9.9, 10). The formula in M3 is based on Equation (3-1) and is presented as follows:

```
=SQRT(SUMXMY2($B3:$E3, H$3:H$6))
```

This formula computes the Euclidean distance between L1 and C1. Pay close attention to the use of absolute references in the formula. The use of absolute references here is for easy autofill. Function SQRT computes the square root of a value, and function SUMXMY2 computes the sum of squares of differences of corresponding values in two arrays (in this case, the two arrays are B3:E3 for L1 and H3:H6 for C1).

Continue the process by following these instructions:

1. Select cell M3 and autofill it horizontally to P3.
2. Select cells M3:P3; autofill them together to cells M52:P52. By far, the distances from each data point to every centroid have been computed. Part of our worksheet looks like Figure 3-4.

	J	K	L	M	N	O	P	Q
1				Distances				
2	C3	C4		C1	C2	C3	C4	Cluster
3	11	15		183.2711	143.9547	93.09672	35.49648	
4	100	200		211.5767	173.0825	123.6851	46.95743	
5	50	80		220.8871	181.7503	128.1015	30.34798	
6	25	40		167.1332	127.7919	77.47258	44.79955	
7				234.6365	195.7625	140.4742	36.41428	
8				171.4611	133.3505	76.57676	31.82766	
9				68.20777	58.21821	72.76675	169.0089	
10				192.6886	153.4684	100.7621	29.15476	
11				253.2969	214.0251	160.8913	60.17475	
12				129.5017	90.53927	36.29049	71.66589	

Figure 3-4. *Distances computed in worksheet 1*

3. Data in column Q specify the clusters to which each data point belongs based on the shortest distance rule. In cell Q3, enter the following formula:

   ```
   =INDEX($M$2:$P$2,1,MATCH(MIN(M3:P3),M3:P3,0))
   ```

   The syntax of function MATCH is MATCH(lookup_value, lookup_array, match_type). If match_type = 0, the function is looking for an exact match in the lookup_array for the lookup_value. If not found, an #N/A error (value not available error) is returned. If found, the function returns the relative position in the array. Note, the first position in Microsoft Office tools is always 1.

   In the preceding formula, the MATCH function finds the relative column position of the minimum value of the array M3:P3 (which is 4 in this specific case). The column position is fed to the INDEX function to find the proper cluster name in the array M2:P2. This formula determines in which cluster the data point L1 is.

As there are only four clusters, a nested IF statement instead of the preceding formula suffices the purpose, too.

```
=IF(MIN(M3:P3)=M3,"C1",IF(MIN(M3:P3)=N3,"C2", IF(MIN(M3:P3)
=O3,"C3","C4")))
```

In the nested IF formula, if the minimum value is the same as M3, return "C1"; else if the minimum value equals N3, return "C2"; else if the minimum value equals O3, return "C3"; else returns "C4".

We can tell when there are too many clusters, INDEX and MATCH functions are a better programming approach than nested IF statements.

4. Autofill Q3 to Q52. By now, our data look like Figure 3-5.

J	K	L	M	N	O	P	Q
			Distances				
C3	C4		C1	C2	C3	C4	Cluster
11	15		183.2711	143.9547	93.09672	35.49648	C4
100	200		211.5767	173.0825	123.6851	46.95743	C4
50	80		220.8871	181.7503	128.1015	30.34798	C4
25	40		167.1332	127.7919	77.47258	44.79955	C4
			234.6365	195.7625	140.4742	36.41428	C4
			171.4611	133.3505	76.57676	31.82766	C4
			68.20777	58.21821	72.76675	169.0089	C2
			192.6886	153.4684	100.7621	29.15476	C4
			253.2969	214.0251	160.8913	60.17475	C4
			129.5017	90.53927	36.29049	71.66589	C3

Figure 3-5. *After first round of computation in worksheet 1*

5. We now need to start the second round of the clustering process. Make a copy of worksheet 1 and rename the new worksheet 2. Yes, the name of the new worksheet is number 2! Also, make sure worksheet 2 is right to worksheet 1. Their order is very important.

   In worksheet 2, we need a formula in cell H3 to recalculate the mean of attribute Murder for all data points inside cluster 1. The needed data are inside the worksheet 1, however. A formula like the following can accomplish the job for us:

```
=AVERAGEIFS('1'!$B$3:'1'!$B$52, '1'!$Q$3:'1'!$Q$52,H$2)
```

Though this formula is inside worksheet 2, the data used are all in worksheet 1, and '1'! references worksheet 1 by its name which is "1" indeed. This formula makes use of the function AVERAGEIFS to compute the mean of attribute Murder for all data points inside cluster 1. Note, only the data points (column B only, too) whose cluster name matches cell H2 (C1) are averaged.

However, the preceding formula is not good enough!

First, this formula is computing an average. In the process of k-means clustering, it is possible that one cluster has no data points. In this case, the denominator will be zero and will result in a divided-by-zero error. Thus, we need to make use of IFERROR function to safeguard such a case.

Second and the most important reason is that, when we need to start another clustering iteration in another worksheet 3, we must go to cell H3 of worksheet 3 to change "1" to "2", because worksheet 3 must make use of data in worksheet 2. We do not want to manually revise our formulas again and again.

To answer this challenge, we need to make use of the functions SHEET, INDIRECT, and ADDRESS and operator &.

SHEET is a very interesting function which returns the sheet number of the referenced sheet. The leftmost worksheet in a workbook is numbered 1, the second leftmost worksheet is numbered 2, and so on. The syntax of SHEET is SHEET(reference), where the argument is usually a cell reference, for example, A1.

Function INDIRECT evaluates a text as a cell reference. Function ADDRESS converts two numbers into a cell reference. Function ADDRESS requires a row number and a column number. In Excel, column A is 1, B is 2, and so on. The operator & can concatenate texts (strings) together.

- In cell F1 of worksheet 2, enter the formula `=SHEET(A1)-1`. You will notice that the value in F1 is 1 which can be used to reference worksheet 1.
- In cells F3 to F6, enter numbers 2, 3, 4, and 5, respectively. These numbers can help to pick the right column when we autofill our formula horizontally. Again, in Excel, column B is column 2, column C is column 3, and so on. These numbers can be used to reference columns B, C, D, and E.

Part of our worksheet 2 looks like Figure 3-6.

	B	C	D	E	F	G	H	I	J	K
1					1			Mean		
2	Murder	Assault	UrbanPop	Rape		Cluster	C1	C2	C3	C4
3	12	193	51	21	2	Murder	2.3	5.6	11	15
4	9	222	39	42	3	Assault	15	53	100	200
5	6	226	72	30	4	UrbanPop	9.9	20	50	80
6	8	177	50	17	5	Rape	10	10	25	40
7	7	235	86	39						
8	5	170	78	37						
9	0	35	75	7						

Figure 3-6. *Set up worksheet 2 for formula automatic updates*

In cell H3, enter the following formula:

```
=IFERROR(AVERAGEIFS(INDIRECT($F$1 & "!" & ADDRESS(3,$F3,1))
: INDIRECT($F$1 & "!" & ADDRESS(52,$F3,1)),INDIRECT($F$1 &
"!$Q$3"):INDIRECT($F$1 & "!$Q$52"),H$2),0)
```

In this formula

- IFERROR makes sure when there is an error, the average is set to 0.
- `ADDRESS(3,$F3,1)` gives `$B$3`.
- `INDIRECT($F$1 & "!" & ADDRESS(3,$F3,1))` ➔ `'1'!$B$3`.
- `INDIRECT($F$1 & "!" & ADDRESS(52,$F3,1))` ➔ `'1'!$B$52`.

- `INDIRECT($F$1 & "!" & ADDRESS(3,$F3,1)) : INDIRECT($F$1 & "!" & ADDRESS(52,$F3,1))` ➔ `'1'!$B$3:'1'!$B$52`.
- `INDIRECT($F$1 & "!$Q$3"):INDIRECT($F$1 & "!$Q$52")` ➔ `'1'!$Q$3:'1'!$Q$52`.

The preceding formula generates a final formula which is the same as

```
=IFERROR(AVERAGEIFS('1'!$B$3:'1'!$B$52,
'1'!$Q$3:'1'!$Q$52,H$2),0)
```

The advantages of the preceding formula are as follows:

- We can autofill from H3 to K6.
- Together with the use of the SHEET function in cell F1, we can automate the referencing to worksheet 2 in worksheet 3 as long as we place worksheet 3 to the immediate right of worksheet 2.

6. Autofill from cell H3 to cell K6. Our worksheet can be compared to Figure 3-7.

F	G	H	I	J	K
1			Mean		
	Cluster	C1	C2	C3	C4
2	Murder	2	2.82	6.692308	9.789474
3	Assault	4.175	45.9	108.2308	215.3684
4	UrbanPop	52.125	55.5	64.84615	65.10526
5	Rape	10.875	12.7	19.69231	26.15789

Figure 3-7. *The centroids in worksheet 2*

7. By now, the distances of each data point to the four new centroids have been automatically computed, together with their belonging clusters. In cells R1 and R2, enter numbers 1 and 2, respectively. Select these two cells and autofill to cell R52. Cells R1:R52 now have numbers 1, 2, 3, …, 52.

8. In cell S2, enter "Old cluster". Column S stores the belonging clusters of all the data points in the previous iteration (such data are in worksheet 1).

9. In cell T2, enter "difference". Column T books the cluster switches between the current iteration and the last iteration. Part of our worksheet looks like Figure 3-8.

M	N	O	P	Q	R	S	T
Distances					1		
C1	C2	C3	C4	Cluster	2	Old cluster	difference
189.3638	147.6883	86.06637	27.03322	C4	3		
220.5397	179.3882	118.8041	31.25793	C4	4		
223.569	181.7076	118.4377	13.77279	C4	5		
173.0506	131.3879	70.41714	42.27735	C4	6		
235.0398	193.3862	129.9646	31.5387	C4	7		
169.8793	128.4613	65.50482	48.63186	C4	8		
38.63239	23.22719	75.31087	181.9163	C2	9		
194.7015	152.8115	90.07965	21.99175	C4	10		
256.712	214.8413	151.5295	44.8096	C4	11		
133.1132	91.68589	31.99575	80.64056	C3	12		
41.39157	30.3101	78.20164	184.4349	C2	13		
50.14042	12.09555	57.995	163.2907	C2	14		
215.8487	173.9986	110.7278	15.42572	C4	15		

Figure 3-8. *Ready to compare with worksheet 1*

10. In cell S3, enter the following formula and autofill to S52:

```
=INDIRECT($F$1 & "!Q" & R3)
```

This formula references the value in cell Q3 of worksheet 1.

11. In cell T3, enter the following formula and autofill to T52:

```
=IF(Q3=S3,0,1)
```

The preceding formula defines that for a given data point, if its last belonging cluster is different from the current belonging cluster, 1 is returned; otherwise, 0 is returned. Our goal is to reach a stable condition when column S has 0s only, that is, all data points stay in the same cluster as they were in the last iteration.

12. Select cells G10 and H10, merge them, and type "convergence" inside. In cell I10, enter the following formula:

```
=SUM(T3:T52)
```

As mentioned earlier, we want to get 0 inside the cell I10. Part of our worksheet 2 should look like Figure 3-9.

F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1			Mean						Distances			1		
	Cluster	C1	C2	C3	C4		C1	C2	C3	C4	Cluster	2	Old cluster	Difference
2	Murder	2	2.82	6.69231	9.78947		189.364	147.688	86.0664	27.0332	C4	3	C4	0
3	Assault	4.175	45.9	108.231	215.368		220.54	179.388	118.804	31.2579	C4	4	C4	0
4	UrbanPop	52.13	55.5	64.8462	65.1053		223.569	181.708	118.438	13.7728	C4	5	C4	0
5	Rape	10.88	12.7	19.6923	26.1579		173.051	131.388	70.4171	42.2773	C4	6	C4	0
							235.04	193.386	129.965	31.5387	C4	7	C4	0
							169.879	128.461	65.5048	48.6319	C4	8	C4	0
							38.6324	23.2272	75.3109	181.916	C2	9	C2	0
	convergence		4				194.701	152.811	90.0796	21.9918	C4	10	C4	0
							256.712	214.841	151.529	44.8096	C4	11	C4	0
							133.113	91.6859	31.9957	80.6406	C3	12	C3	0
							41.3916	30.3101	78.2016	184.435	C2	13	C2	0
							50.1404	12.0956	57.995	163.291	C2	14	C2	0

Figure 3-9. *After the second round in worksheet 2*

13. As there are four data points that have switched clusters in the second iteration, we need continue the clustering process. Make a copy of worksheet 2 and give the new worksheet the name 3. Make sure that the worksheet 3 is immediately right to worksheet 2. We shall notice that worksheet 3 is automatically done and part of it looks like Figure 3-10.

F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
2			Mean						Distances			1		
	Cluster	C1	C2	C3	C4		C1	C2	C3	C4	Cluster	2	Old cluster	Difference
2	Murder	2	2.65455	7.66667	9.8125		189.364	146.244	73.5445	35.7077	C4	3	C4	0
3	Assault	4.175	47.3636	121.333	226.063		220.54	177.946	106.85	28.2629	C4	4	C4	0
4	UrbanPop	52.13	57.0909	66.8667	62.75		223.569	180.026	105.327	10.3761	C4	5	C4	0
5	Rape	10.88	13.7273	19.5333	27.25		173.051	129.981	58.2219	51.7498	C4	6	C4	0
							235.04	191.574	116.9	27.6842	C4	7	C4	0
							169.879	126.586	52.9584	59.1083	C3	8	C4	1
							38.6324	22.9325	87.9515	192.773	C2	9	C2	0
	convergence		4				194.701	151.2	76.9519	31.9987	C4	10	C4	0
							256.712	213.185	138.351	35.0594	C4	11	C4	0
							133.113	90.2234	22.58	90.9081	C3	12	C3	0
							41.3916	29.5418	90.6807	195.343	C2	13	C2	0
							50.1404	12.4859	71.0211	173.731	C2	14	C2	0

Figure 3-10. *K-means clustering after three rounds in worksheet 3*

14. Repeat the preceding clustering process: copy worksheet 3 to worksheet 4; make sure that the new worksheet is named 4 and is right to worksheet 3. Part of worksheet 4 looks like Figure 3-11.

F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
3			Mean						Distances			1		
	Cluster	C1	C2	C3	C4		C1	C2	C3	C4	Cluster	2	Old cluster	Difference
2	Murder	1.78889	3.34167	7.78571	10.133		186.873	139.068	64.9344	38.7852	C4	3	C4	0
3	Assault	6.71111	54.5	130.714	229.8		218.068	170.746	98.4446	28.5673	C4	4	C4	0
4	UrbanPop	51.2222	58.4167	68.8571	61.733		221.187	172.647	95.8334	12.1856	C4	5	C4	0
5	Rape	10.3333	15.75	20.4286	26.6		170.537	122.883	50.0975	54.9747	C3	6	C4	1
							232.754	184.106	107.308	27.9193	C4	7	C4	0
							167.636	119.072	43.6959	63.0488	C3	8	C3	0
							37.1477	27.2578	97.1592	196.494	C2	9	C2	0
	convergence		2				192.278	143.97	67.6201	35.3802	C4	10	C4	0
							254.325	205.814	128.851	31.9826	C4	11	C4	0
							130.681	82.9904	19.414	94.4649	C3	12	C3	0
							40.5161	32.6609	99.6833	199.133	C2	13	C2	0
							47.5399	12.4828	80.5883	177.309	C2	14	C2	0

Figure 3-11. *K-means clustering after four rounds in worksheet 4*

15. Repeat the clustering process again: copy worksheet 4 to worksheet 5; make sure that the new worksheet is named 5 and is right to worksheet 4. The convergence becomes 0 as shown in Figure 3-12.

F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
4			Mean						Distances			1		
	Cluster	C1	C2	C3	C4		C1	C2	C3	C4	Cluster	2	Old cluster	Difference
2	Murder	2.31	3.00909	7.8	10.2857		184.881	136.489	61.6318	42.6894	C4	3	C4	0
3	Assault	8.64	57.0909	134	233.571		216.104	168.195	95.2569	30.1276	C4	4	C4	0
4	UrbanPop	52.6	57.8182	67.6	62.5714		219.063	170.153	92.8412	13.1133	C4	5	C4	0
5	Rape	11.2	15.4545	20.2	27.2857		168.576	120.277	46.7577	58.9015	C3	6	C3	0
							230.541	181.704	104.566	26.4378	C4	7	C4	0
							165.394	116.722	41.3362	66.3451	C3	8	C3	0
							34.9226	29.3897	100.256	200.256	C2	9	C2	0
	convergence		0				190.203	141.439	64.5669	38.8711	C4	10	C4	0
							252.202	203.324	125.853	28.1418	C4	11	C4	0
							128.631	80.4817	17.7336	98.3105	C3	12	C3	0
							38.091	34.938	102.901	202.836	C2	13	C2	0
							45.7697	12.1481	83.1894	181.184	C2	14	C2	0

Figure 3-12. *K-means clustering after five rounds in worksheet 5*

16. Once the convergence has reached 0, no matter how many more iterations we continue, no data will change again. However, it is necessary to confirm it by creating another worksheet 6 to repeat the preceding process.

    Note, it is very important to arrange the six worksheets in order; otherwise, you won't get the correct result. Take a look at Figure 3-13.

	B	C	D	E	F	G	H	I	J
1					5			Mean	
2	Murder	Assault	UrbanPop	Rape		Cluster	C1	C2	C3
3	12	193	51	21	2	Murder	2.31	3.009091	7.8
4	9	222	39	42	3	Assault	8.64	57.09091	133.8
5	6	226	72	30	4	UrbanPop	52.6	57.81818	67.6
6	8	177	50	17	5	Rape	11.2	15.45455	20.2
7	7	235	86	39					
8	5	170	78	37					
9	0	35	75	7					
10	4	198	70	15		convergence		0	
11	13	259	74	30					
12	15	136	52	24					

1 2 3 4 5 6

Figure 3-13. *The six worksheets must be placed in order*

By now, all data points have been successfully clustered.

From the preceding clustering process, we can tell that clustering in Excel is not difficult. Instead, it allows us to go through every single step and therefore develop a solid understanding of how k-means clustering works. I would like to challenge you to examine the numbers inside the four clusters, respectively. I am sure that you will find out some patterns. When you do so, remember that clustering is to group similar records together.

The complete result is in Chapter3-1b.xlsx (it also contains my observation on patterns).

Reinforcement Exercises

The file Chapter3-HW.xlsx provides an exercise opportunity for you. In this exercise, you need go over 15 worksheets to reach convergence. Should you find the exercise difficult, please refer to Chapter3-HW-withAnswers.xlsx.

The file Chapter3-HW-withAnswers-NameManager.xlsx makes use of Name Manager, an Excel built-in feature, to make the formulas simpler.

So far we employ the SHEET function to help automate the k-means clustering process. One critical requirement of the use of SHEET function is to place all worksheets in order. If we do not want to place all the worksheets in order, we can make use of the function CELL instead of SHEET. The file Chapter3-HW-withAnswers-Cell-Function.xlsx shows such a solution. Feel free to explore it.

Excel has some options to configure formula calculations. By default, the "Enable iterative calculation" option is disabled. If we enable iterative calculation in Excel, we can then complete the k-means clustering process in only two worksheets because the second worksheet can perform a loop-like procedure to reach convergence! Due to the space limit of this book, I won't explain how to achieve it. But please be advised that the clustering result obtained based on the preceding worksheet-by-worksheet procedure may be different from what can be acquired through this iterative calculation, though both results are correct, which indicates that there can be more than one optimal clustering solution.

Review Points

For Chapter 3, the following knowledge, skills, and functions are worth reviewing:

1. Centroid, Euclidean distance
2. Reference a worksheet by its name followed by the exclamation symbol "!"
3. Functions INDEX, IF, SQRT, SUMXMY2, and MATCH
4. Functions IFERROR, SHEET, INDIRECT, ADDRESS, and AVERAGEIFS
5. Operator &
6. Worksheet copying to start a new iteration
7. The clustering process